5. Functional Specification

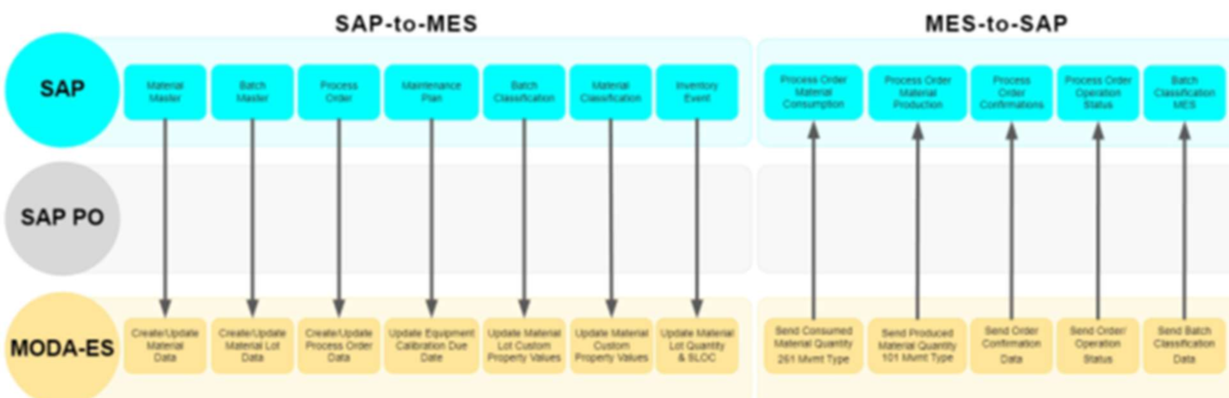
1. Functional Description

The integration must support a **one-to-many** relationship between an SAP Process Order Operation and corresponding MES Orders.

- A single **SAP Process Order Operation/Phase** (e.g., Operation Phase 10 in the master recipe) can have **multiple MES Orders** generated and executed in the MES system.
- Each MES Order can be further modeled as one or more **MES Operations**, depending on shop floor requirements.

The integration shall enable MES to update the status of the SAP operation based on the **collective progress** of its linked MES Orders.

2. Flow Diagram à TO BE REFINED ACCORDINGLY



* One (1) REST API URL endpoint configured for each message type.

1. **Operation Creation & Dispatch**
   - When a process order is released in S/4HANA, the system sends a **Process Order message** to MES.
   - The message contains:
     - Process Order Number
     - Operation Number / Phase
     - Operation Details (work center, duration, etc.)
     - Unique Operation ID (for linking MES Orders back to SAP)
   - MES creates **multiple MES Orders** for a single SAP operation.
2. **Operation Status Updates from MES**
   - **Start Event**:
     - When *any* MES Order linked to the SAP operation is **started**, MES triggers a **Status Update Message** to S/4.
     - S/4 updates the corresponding SAP operation status to **"Started" (system status CNF/PCNF depending on setup)**.
   - **Completion Event**:

- When **all** MES Orders linked to the SAP operation are **completed**, MES triggers a final status message.
- S/4 updates the SAP operation to **“Completed” (system status CNF)**.
- Optional: Intermediate updates (e.g., partial confirmations or yield postings) can also be sent per MES Order if required.

3. Unit Testing

<<<Provide step by step example for unit testing >>>

Test Case	Precondition	Expected Result
Single MES Order Start	1 MES Order linked	SAP operation shows “Started” after message received
Multiple MES Orders, partial completion	3 MES Orders linked; 2 completed	SAP operation remains “Started”
All MES Orders Completed	3 MES Orders linked; all completed	SAP operation shows “Completed”
Error Retry	MES sends duplicate completion message	SAP ignores duplicate and keeps operation completed

6. Design Specification

1. Configuration

<<< Rename Section 5.1.1 “Configuration reference” with the configuration node and repeat it for each unique configuration item.

- *Specify the SPRO configuration required for this custom development >>>*

1. Configuration reference

Configuration path (IMG, table, ...)	

1. Purpose of configuration

<<< Describe here the purpose and relevant information related to the configuration. >>>

2. Workflow

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. Flow Diagram

<<< Please provide the technical process flow diagram >>>

3. Steps Description

<<< Process steps should be descriptive in nature. The aim of the process step is to describe the overall technical process >>>

4. Technical Details

Trigger Mechanism	Mention the start condition for the workflow, e.g. on creation of a purchase document, batch program etc.
Start Condition	Example – The workflow should start only for certain document type, workflow should start only if credit amount is greater than 250000 etc.
Business Object	Mention the business object, if possible. Otherwise indicate the object in general terms (e.g., Purchase Requisition)
Standard Workflow Task / Template	In case of enhancement required for delivery workflow is required.

Level of Approval Required	
Agent Determination Technique	Role - Security Role Org Unit - HR Org Structure Custom Table - Agents in custom table Distribution Lists Unspecified - To be decided in Functional Specification Other <use to elaborate on a selection of "Other"> If the agent determination technique is different for each foreground step then please repeat this section.
Mention Logic for Agent Determination (if any)	
Notification Destination	Internal User (Mail Inbox) External User (email address)
Workflow Notifications Text	If any specific work item text/work item subject to be used.
Escalation Handling (if any)	If any deadline monitoring is to be done. Example: If approver does not approve for 3 business days notify his supervisor.
Integration with Portal	
Configuration Dependencies	Example setting up a new organization structure.

Error Handling (if any)	An exception situation could occur if workflow routes to a one position is vacant/not available (i.e. no user is assigned to that position.) If a specific report or additional information is required. Add attachment if necessary.
Substitution	

5. **Authorization**

<<< Explain which roles should be added or used to approve/reject and execute workflow items. Enter any custom authorization required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

3. **Report**

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. **Selection Screen Details**

<<< The functional designer should be able to detail exactly what he/she wants at the selection screen merely by using this table. The programmer will be able to construct the screen directly from the details in this table. Some technical knowledge will be needed for the complete production of this table>>>

Name	Type	Parameter or Select Option	Comments (Range, Single/Multiples selections, Patterns Mandatory etc.)	Default Value
	Table-Field Check Box Radio Button with Group	Parameter Select Option		
	Table-Field Check Box Radio Button with Group	Parameter Select Option		

3. **Desired Screen Design**

<<< Enter attachment if necessary >>>

4. **Technical Details**

<<< Information like relevant database tables, data retrieval logic, type of report like (simple list report or ALV), sorting order, detail functionality, other display attributes, special interaction on clicking one or more columns etc. can mentioned here >>>

5. Starting Conditions

<<< When should the report be run? Does an interface need to be run before the report is valid, and (more commonly), should it be a batch only program (with added security) or is it needed on-line as well?

E.g. 'This program will be run after month-end billing.

E.g. 'This program will be run each time a sales order is saved >>>

6. Data Mapping Tables

<<< List of all the fields along with their details are to be mentioned here. Look and feel wise, a desired report design can also be specified here >>>

Field Name	Field Description	Output Length	Output Type	Format	Position	Screen No / Field Name

7. Report Example

<<< Use Attachment if necessary >>>

8. Authorization

<<< Explain which roles should be added or used for these reports. Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

4. **Interface**

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. **Technical Details**

Interface Name	Material master Interface
Direction (with respect to this system)	Inbound Outbound other If other, please specify exactly
Interface Type	Batch near real-time real-time other If other, please specify exactly
Interface Frequency	Hourly Details: Daily Details: Weekly Details: Monthly Details:

	Quarterly Details: Yearly Details: On-Demand Details: Other Details:
Type of Records Sent	Delta Fields Delta full record other If other, please specify exactly
Volume (per single execution)	Average Volume: <Volume> records per interface execution Peak Volume: <Lower Volume – Upper Volume>

3. **Flow logic**

<<< Please explain any flow logic, calculations, rules, etc.. that should be implemented in this interface >>>

PROPOSAL à DEPENDS ON THE TECHNICAL TEAM SOLUTION

- **Key Design Considerations**
- **Mapping Table / Relationship Logic**
  - A table link will maintain the mapping between:
    - SAP Process Order Number
    - SAP Operation Number/Phase
    - MES Order Numbers
  - This ensures that S/4 knows which MES Orders belong to which SAP operation.
  - The MES system will reference this ID in all status updates.

- **Status Logic**

Event in MES	Condition	Action in S/4
Any MES Order Started	First MES order changes to *Started*	Set SAP Operation Status = **Started**
All MES Orders Completed	All MES orders under this operation = *Completed*	Set SAP Operation Status = **Completed/Confirmed**

- **Message Types**
    - Outbound: **Process Order Dispatch Message** (S/4 → MES)
        - JSON structure similar to current IDoc PROCESS_ORDER but enhanced for multiple MES Orders.
    - Inbound: **Operation Status Update Message** (MES → S/4)
        - JSON fields:
        - Process Order ID
        - Operation/Phase ID
        - MES Order ID
        - Status (Started/Completed)
        - Timestamp
- **Error Handling & Reprocessing**
    - If a status update fails (e.g., due to network errors), MES will re-send the message.
- **Change Pointers**
    - Change pointers will not be triggered by MES updates but must be active for outbound Process Order messages to ensure delta updates when SAP makes changes (e.g., operation rescheduling).

4. Interface Data layout

<<< Please list the source and destination data elements, plus any mapping that will be required for this interface. If IDOC, include segment name in structure column. Excel matching this format can be attached in place of this table. >>>

Source Structure	**Source Field**	**Description**	**Data Type**	**Length**	**Transformation**	**Target Structure**	**Target Field**	**Description**	**Data Type**	**Length**	**Mand / Opt.**	**Comments/Remarks**

SAP Source Field	**SAP Table**	**JSON Field (Outbound to MES)**	**Direction**	**Notes**
Process Order Number	AFKO-AUFNR	processOrder	Outbound	Unique process order ID
Operation Number	AFVC-VORNR	operation	Outbound	Links MES Order to SAP Operation
Phase	AFVC-APLZL	phase	Outbound	Optional if you use phase reporting
Operation Status	JEST-STAT	status	Inbound/Outbound	MES sends STARTED/COMPLETED
MES Order ID	If needed; TBD	mesOrder	Inbound	Maintains MES ↔ SAP relationship

Confirmation Date/Time	AFRU-ISDD/AFRU-ISDZ	timestamp	Inbound	For status updates
Plant	AFKO-WERKS	plant	Outbound	Required for MES routing
Work Center	AFVC-ARBPL	workCenter	Outbound	Shop floor execution

Possible link table that we can create to connect both data in SAP and MES à PROPOSAL DEPENDS ON THE TECH

Source	Field Description	Target JSON Field	Remarks / Business Rule
Z-Interface Table	MES Order ID	mesOrderId	Unique MES sub-order per operation/phase
MES System	Start Timestamp	mesStartTime	When MES marks a sub-order as started
MES System	Completion Timestamp	mesEndTime	When MES marks a sub-order as complete
MES System	Completion Flag	completionIndicator	Boolean to trigger SAP confirmation

5. Mapping Rules & Conversion Criteria

<<< This section should contain any additional mapping rules and conversion criteria not covered in the previous section. >>>

6. Special Case: Bi-Directional Real-Time Interface

<<< If you know this interface will be a bi-directional real-time interface (i.e. the “Source” system sends and receives data in the same execution), then a second data mapping is required. If applicable, duplicate the table from Section 4.4.4 and capture the “return data” mapping rules for the “Source” system >>>

7. Sample Data

<<< Please provide two attachments of sample source data with the expected target data after this interface is executed. Please supply the sample data in the native format or .csv, and preferably zipped >>>

8. Data Retention

<<< In file based interfaces a “backup” copy of interface data can be retained in the middleware for each execution. This can be useful for reconciliation purposes. Please indicate the retention period for this interface. If not file based, then the source or target system must fill any data retention requirements >>>

Selection		**Comments**
	None	
	7 Days	
	15 Days	
	30 Days	
	Other	

9. Middleware Solution

<<< This section should contain an outline of the chosen middleware solution and the processes involved. Middleware specific configuration should be specified >>>

10. Interface Scheduling

<<< Please describe any requirements around the timing of this interface >>>

11. Authorization

<<< Explain which roles should be created / added or users / IT for reprocessing errors or ad hoc requests. Enter any custom authorization if required. Enter the file path or folder structure to which users/IT will need access to >>>

No	Business Catalog	Authorization Parameter	Parameter Value

12. Other system documentation

<<< Reference the other system's documentation, when relevant >>>

5. Conversion

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. **Technical Details**

Conversion Name	

3. **Conversion Data Layout**

<<< Please list the source and destination data elements, plus any mapping that will be required for this conversion. If uploading from file, source structure can be omitted. Excel matching this format can be attached in place of this table. >>>

Source Structure	Source Field	Description	Data Type	Length	Transformation	Target Structure	Target Field	Description	Data Type	Length	Mand / Opt.	Comments/Remarks

4. **Mapping Rules & Conversion Criteria**

<<< This section should contain any additional mapping rules and conversion criteria not covered in the previous section. >>>

5. Sample Data

<<< Please provide two attachments of sample source data with the expected target data after this conversion is executed. Please supply the sample data in the native format or .csv, and preferably zipped >>>

6. Authorization

<<< Explain which roles should be created/added or used for loading data. Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

6. Enhancement

1. Business Add-Ins (BADIs)

BADI Property	Value/Object
System	*<<< BTP, S/4 HANA, ...>>>*
Transaction	
Enhancement Spot	
BADI Name	
Enhancement Implementation	
BADI Implementation	

Class	
Method	
Filter	
OData Service	

2. **Implicit Enhancement**

Property	**Value/Object**
Transaction	
Enhanced Object	
Implementation	

3. **User-Exits**

Property	**Value/Object**
Transaction	
Main Program	
Includes	
Form Routines	

4. **CDS Views Extension**

1. **Technical Reference**

Property	Value/Object
Original CDS View Name	
Extended CDS View Name	
Purpose of Extension	
Extension Type	☐CDS View Extension ☐Custom CDS consuming Standard CDS ☐View with Additional Associations or Joins ☐Metadata Extension
Odata Exposure	☐Yes ☐No
Input Field Parameters	
Service Definition	
Service Binding	

2. **Fields Added**

Field Name	Data Element	Source Table	Description	Annotations

5. **Function Exits**

Property	Value/Object
Transaction	
Enhancement	
Function Module Name	
Includes	

6. **Field Exits**

Property	Value/Object
Enhancement	
Main Program Name	
Function Module Name	
Field Exit Id	
Screen Number	
Screen Field Name	
Conditions for execution	

7. **Menu Exits**

Property	Value/Object
Enhancement	
Menu/Path	
Function/Transaction Code	

8. **Screen Exits**

Property	Value/Object
Enhancement	
Main Program Name	
Screen Number	
Program Name & Sub-Screen Number	

9. **Search Help Exits**

Field Name	Field Description	Import / Export (I/E)	Key Field (Y/N)	Data Element	Type (CHAR, NUMC)	Length	Default Value

10. **Search Help assignment**

Property	Value/Object
Standard Search Help	
Collective Search Help	
Elementary Search Help	

11. Business Transaction Events (BTE)

Property	Value/Object
Transaction	
BTE Number	
Product Name	
Function Module	

12. Custom Transaction

<<< Functional details of custom transaction can be incorporated here. Number of screens required and flow diagram can be included and provide the selection screen shot along with the table name and field name and screen shot for the required output >>>

13. Requirement routine

Menu/Submenu	

Routine number	
Business logic required	

14. Substitution

Validation Description	Fields required for validation	Point of Validation	Table used in validation	Business Rules

Substituted Field	Derived from Field	Table used in Substitution	Business Rules

15. Flow logic

<<< Please explain any flow logic, calculations, rules, etc that should be implemented in this enhancement >>>

16. Authorization

<<< Which authorization object should be used for controlled execution? Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

7. **Form**

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. **Form Layout**

<<< Refer to the following for an output samples for Window mapping, Label Description and Field mapping >>>

3. **Layout Windows**

Reference	Print on page	Label Position

		X : Y :
		X : Y :
		X : Y :
		X : Y :
		X : Y :
		X : Y :
		X : Y :

4. Field Mapping

Field	Field Description	Functionality	Logic	Print on page	Font	Font Format	Window

5. Standard Texts / Text Modules

Reference	Text	Print on page	Label Position	Font	Output Format	Font Format

6. **Translation**

Reference	Description of use (in Language1)	Description of use (in Language2)	Description of use (in Language3)	Text Module Name	Notes

7. **Layout Details**

Position of Left Margin (specify unit)	
Position of Right Margin (specify unit)	
Position of Logo (specify unit)	
Logo (specify logo)	
Position of Main Window (specify unit)	

8. **Flow logic**

<<< Please explain any flow logic, calculations, rules, etc that should be implemented in this form >>>

9. Authorization

<<< Explain which roles should be created/added or used for printing and testing forms. Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

8. Fiori Application

1. Header Information

Application Title	
Application ID	
Type of Enhancement	☐Custom Application ☐ Standard Application
Development Type	<<<*Fiori Elements App ....Free Style UI5 App*>>>
Application Type	<<<*List Report, Object Page , Over view Page ,etc >>>*
UI Enhancements	☐Custom Fields Added ☐UI Layout Modified

	☐Extensibility Hook Used ☐Fragments or Views Introduced

2. Technical Reference

Object Name	Object Type	Object Description
<<< *Odata Object* >>>		
<<< *CDS View* >>>		
<<< *Custom Fields* >>>		
<<< *Catalogs* >>>		
<<< *Rules* >>>		

3. Desired Screen Design

<<< *Enter attachment if necessary* >>>

4. Technical Details

<<< *Information like relevant database tables,CDS Views,ODATA services , data retrieval logic, detail functionality, other display attributes, special interaction on clicking one or more columns etc. can mentioned here* >>>

5. Authorization

<<< *Enter Authorization Objects/fields, to be used and specific user Groups* >>>

No	Business Catalog	Name of Space (L2)	Name of Page (L3)	Name of Section (L4)	Name of App/Tile(L5)	Authorization Parameter	Parameter Value

7. Custom Tables/Structure

<<< This section should detail the attributes of any new custom table created for one of the above sections, and the properties of its fields.

NB: Existing Data Elements and/or Domains should be used whenever possible when creating custom table fields, in order to avoid unnecessary typos. In this instance, the data table row for that field should not be completed beyond 'Domain', as the remaining attributes will be default values for the selected Domain. >>>

Table Name	
Short text	
Size category	
Table maintenance allowed	
Maintenance Type	Manual / Automatic Maintenance (application table) Transportable Maintenance (customizing table)
Data class	
Buffering	

Table maintenance generation								
Authorization Group								
Change Log Enabled (Y/N) *(mandatory for GxP related table)*								
SPRO Path *(mandatory for customizing tables)*								
Field Name	**Data Element**	**Domain**	**Type**	**Length**	**Check Table-Field**	**Key Field**	**Foreign Key**	**Description**
Comments								

8. Error Handling

<<< Provide Error Handling details here. Job run notifications, error notifications, E-Mail messaging, custom programming, etc. may be required >>>

1. Error Messages

<<< Describe the expected error messages for different error conditions >>>

Error Message Number	Error Message Text (70 characters)	Error Conditions

9. **Validation**

1. **Test Case References**

<<< List the Test Case(s) used to validate the functionality / configuration covered in this document (IQ / OQ). >>>

Test Case ID	Test Case	Comment